

Session 5 — Distances, and how anyone knows

Practical Astronomy Intensive

9 questions 15 minutes 6 concepts 9 diagnostic

They arrived knowing Sirius is 8.6 light years away. This checks whether the session turned that fact back into a measurement — parallax as geometry, the magnitude scale running backwards, and $1/d^2$ quartering rather than halving.

- 1 A student writes: "Proxima Centauri is 4.2 light years away, so its light is 4.2 years old." The second half is right. What is a light year?

lightyear-scale Recall Multiple choice

- a The distance Earth travels along its orbit in one year
REVEALS Light-year confused with the AU / Earth's orbital path
- b The speed at which light travels over interstellar distances
REVEALS Light-year read as a speed
- c A length of time — the year light takes to arrive from a star
REVEALS Light-year read as a unit of time
- d A distance — the 9.46 trillion km light covers in one year ✓

The name does the damage. It is a distance, and the fact that light takes a year to cross it is what defines the distance, not what the unit measures. Neptune sits about 4 light hours out; Proxima is roughly 7,000 times farther.

- 2 You are told 61 Cygni lies 11.4 light years away, measured by parallax. What kind of claim is that?

stellar-parallax Recall Multiple choice

- a A theory about stellar distances that fits the observations but could be replaced by a better one
REVEALS Parallax as theory or inference rather than direct geometry
- b A figure worked back from the star's colour and spectral type
REVEALS Parallax confused with spectroscopic distance; ignores the geometry entirely
- c A direct geometric measurement: measure the angular shift, take the 2 AU baseline as the other side of the triangle, and the distance follows ✓
- d An estimate obtained by assuming 61 Cygni is as luminous as the Sun and comparing brightness
REVEALS Parallax confused with brightness-based distance — it assumes nothing about the star

Parallax assumes nothing about the star itself — only that Earth's orbit is 2 AU across and that angles can be measured. It is the one rung of the distance ladder that is pure trigonometry, which is why everything above it is calibrated against it.

- 3 Voyager 1 is leaving the solar system at about 17 km/s — the fastest thing we have ever launched. Roughly how long would it need to cover the 4.2 light years to Proxima Centauri?

lightyear-scale Apply Multiple choice

- a About 4.2 years, since that is how long light takes and Voyager is fast
REVEALS Light-year read as a unit of time, applied to any traveller
- b About 75,000 years ✓
- c About 400 years, well within a few generations of a crewed mission
REVEALS 4.2 light years believed to be nearly within reach
- d About 40 years — roughly ten times its journey to the edge of the solar system so far
REVEALS Interstellar space imagined as the solar system scaled up a little

Nine trillion km per light year, times 4.2, divided by 17 km/s, is roughly 75,000 years. The point is not the arithmetic but what it does to the word 'nearest': the closest star is a gap no spacecraft we can build meaningfully crosses.

- 4 Two identical stars sit in the same field of view, but one is exactly twice as far away as the other. How much fainter does the far one look?

inverse-square Apply Multiple choice

- a Four times fainter — the same light is spread over four times the area ✓
- b It looks the same; distance changes the size of a star's disc, not its brightness
REVEALS Brightness thought to be independent of distance
- c Twice as faint — double the distance, half the brightness
REVEALS Brightness assumed to halve when distance doubles — the linear intuition
- d Eight times fainter — light spreads through a volume, so $1/d^3$
REVEALS Inverse-cube error: light spreads over a surface, not a volume

Flux falls as $1/d^2$. At twice the distance the star's light crosses a sphere with four times the area, so a quarter reaches you. The 'half' answer is the commonest wrong one, and it wrecks every distance calculation downstream.

- 5 Sirius (apparent magnitude -1.5) looks far brighter than Rigel (apparent magnitude $+0.1$). Which star actually puts out more energy?

luminosity-absmag Apply Multiple choice

- a They are equal; magnitude already accounts for distance
REVEALS Apparent magnitude confused with absolute magnitude
- b Rigel, because its magnitude number is larger and larger numbers mean more power
REVEALS Magnitude scale read forwards — bigger numbers are fainter, not brighter
- c Rigel, by an enormous margin — Sirius only wins the sky because it is about 8.6 ly away while Rigel is roughly 860 ly ✓
- d Sirius — it is the brightest star in the night sky, so it must be the most powerful
REVEALS Brightest-looking star assumed to be the most powerful — apparent magnitude read as true output

Apparent magnitude mixes true output with distance and tells you neither on its own. Sirius is a modest A1 star next door; Rigel is a blue supergiant roughly 100 times farther out and tens of thousands of times more luminous.

6 Betelgeuse is distinctly orange-red; Rigel is blue-white. Both are in Orion, both are visible tonight. What does that colour difference tell you?

star-colors-temp

Apply

Multiple choice

a Betelgeuse is much hotter — red-hot is hotter than blue, as anyone who has seen a stove ring knows

REVEALS Red = hot, blue = cool — the stove-ring intuition, exactly backwards for stars

b Betelgeuse is closer, and nearer stars look redder

REVEALS Colour confused with distance or reddening effects

c Rigel is much hotter — blue-white is above 10,000 K, red is around 3,000 K ✓

d They are made of different elements, and colour reveals composition

REVEALS Colour read as composition rather than temperature

Colour is a thermometer: hotter blackbodies peak at shorter wavelengths, so hotter means bluer. Composition needs the absorption lines, not the tint. Worth adding at the eyepiece: the eye's colour cells barely fire at these brightnesses, so only the brightest stars show any tint at all.

7 You measure the same star twice on one clear night: once near the horizon, once near the zenith. The horizon reading is several tenths of a magnitude fainter. What happened?

atmospheric-extinction

Apply

Multiple choice

a The night was clear, so extinction is zero; the difference must be instrument drift

REVEALS 'Clear' treated as extinction-free — even the zenith costs a few tenths of a magnitude

b The star was farther away when near the horizon, and light falls off with distance

REVEALS Horizon dimming attributed to distance — the same error that makes people think the setting Sun has moved away

c Near the horizon its light crossed several times more atmosphere, which dimmed and reddened it — the star itself did nothing ✓

d Nothing physical — a star's brightness is fixed wherever it sits, so one of the two readings must be a mistake

REVEALS Brightness assumed fixed regardless of altitude — airmass ignored

Airmass is roughly $\sec(z)$: at 60° from the zenith you are looking through twice the air, and near the horizon through several times more. It is the same physics that turns the setting Sun orange — the long slant path scatters the blue away. Observe high, image high.

8 A student has a photometer reading for a star and wants its absolute magnitude tonight. What else does she need, and why?

luminosity-absmag

Reason

Multiple choice

- a The star's colour, which converts apparent magnitude to absolute magnitude
REVEALS Absolute magnitude thought to follow from colour rather than distance
- b Nothing — she can subtract 5 from the apparent magnitude, since the scale is linear
REVEALS Magnitude scale treated as linear rather than logarithmic
- c** The distance — absolute magnitude is a calculation, the brightness the star would show at 10 parsecs, and you cannot get there without knowing how far away it is ✓
- d Nothing more; absolute magnitude is a direct measurement, just made with a better-calibrated instrument
REVEALS Absolute magnitude believed to be measurable rather than computed from distance

$m - M$ is the distance modulus: absolute magnitude falls out only once distance is in hand. This is exactly why parallax has to be taught before magnitude — the geometry comes first, the comparison second. And the scale is logarithmic: five magnitudes is a factor of 100 in brightness, not five of anything.

9 The Greeks argued that if Earth orbited the Sun, the stars would visibly shift over the year — they saw no shift, so Earth must be still. Their observation was honest. Where did the argument fail?

stellar-parallax

Reason

Multiple choice

- a The stars do not shift at all; parallax is an instrumental artefact of modern telescopes
REVEALS Parallax treated as an artefact rather than a real annual shift
- b The shift is large but the Greeks had no way to record star positions accurately
REVEALS Expecting stellar parallax to be a visible shift — it is sub-arcsecond
- c** The stars are so far away that even the largest stellar parallax is under one arcsecond — far below what an unaided eye can detect ✓
- d Parallax only appears for stars on the ecliptic, and the Greeks watched the wrong stars
REVEALS Believing parallax is limited to particular sky regions rather than shrinking with distance

The logic was sound and the data were real; the missing piece was scale. Bessel needed 1838 and a heliometer to pull 0.3 arcseconds out of 61 Cygni. The absence of a naked-eye shift is exactly what enormous stellar distances predict.